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Cloning and Characterization of New Repetitive Sequences in Field Bean (*Vicia faba* L.)

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Five new repetitive sequences have been isolated from the *Vicia faba* genome, by cloning bands visible on agarose gel electrophoresis after digestion of genomic DNA with various restriction enzymes. The sequences were 109 to 584 bp long, their abundance ranging from 5×10^4 to 5×10^5 copies per haploid genome. Southern blot and *in situ* hybridization revealed that four of five newly isolated repeats were dispersed in the *V. faba* genome. One of the repeats (TIII15) showed tandem organization with several major hybridization spots on mitotic chromosomes *in situ*. These sites were distributed in euchromatic as well as in heterochromatic chromosomal regions, and in several loci they were simultaneously localized with previously described *Fok*I repeated elements. The sequence of TIII15 comprises four 26–27 bp subrepeats, but shares no homology to *Fok*I elements which have similar sequence organization. All newly described sequences were highly specific for *V. faba*, with little or no hybridization to DNA of other *Vicia* species, and no hybridization to DNA of other legumes tested. © 1999 Annals of Botany Company

Key words: *Vicia faba*, field bean, repeated DNA sequences, FISH, PRINS, genome organization, copy number.

INTRODUCTION

Repetitive DNA sequences represent a considerable portion of higher plant genomes (Flavell *et al.*, 1974). However, relatively little is known about evolutionary relationships and functional significance of repetitive elements within the genomes of individual plant species. To address these questions, isolation and thorough characterization of corresponding sequences is required, as well as studies of their distribution within evolutionary-related species. Field bean (*Vicia faba*) is a suitable model for these types of studies. It represents one of the largest legume genomes ($1C = 13{\cdot}5$ pg; Bennett and Leitch, 1995), having a high proportion of repetitive DNA (> 85%; Flavell *et al.*, 1974), part of which has already been characterized in detail. Beyond rDNA genes, three families of highly repetitive DNA elements have been isolated so far. The most abundant repeat, the *Fok*I element (Kato, Yakura and Tanifuji, 1984), is present at about 10^7 copies per haploid genome (4·5% of *V. faba* genome; Kato *et al.*, 1984; Maggini *et al.*, 1995). *Fok*I repeats are arranged in tandem, individual elements being 59 bp long, and are concentrated at a limited number of genomic loci. Visualization of these loci by *in situ* hybridization on metaphase chromosomes revealed several bands, which corresponded with some of the heterochromatic chromosomal regions (Fuchs *et al.*, 1994).

The two other families represent dispersed repeats. The *Bam*HI family includes seven classes of repeats 250 to 1750 bp long, which share partial sequence homology. Each of the classes comprise about 3% of the genome (Kato *et al.*, 1985). *Ty1-copia* retrotransposons have been detected in the field bean genome by PCR amplification using primers derived from conserved regions of the reverse transcriptase gene (Flavell, Smith and Kumar, 1992). The isolated 250 bp fragment was estimated to comprise about 2% of the genome (Pearce *et al.*, 1996). However, if all of these fragments represent parts of full length copies of *Ty1-copia* elements, this retrotransposon would comprise 40% of the field bean genome (Pearce *et al.*, 1996).

In this study we focused on the isolation of new families of highly repetitive DNA from the genome of field bean. Newly isolated repeats were characterized with respect to their copy number, genomic organization and occurrence in other legume species.

MATERIALS AND METHODS

Plant material

Seeds of field bean (*Vicia faba* ssp. *faba* var. *equina* Pers., $2n = 12$) 'Inovec' with the standard karyotype were obtained from Dr M. Vavák (Horná Streda, Slovakia). Seeds of field bean line ACB with reconstructed karyotype were kindly provided by Dr I. Schubert (IPK Gatersleben, Germany); for a detailed description of chromosome translocations in this line see Schubert, Rieger and Michaelis (1986). Seeds of the other *Vicia* species were received from the legume gene bank at Agritec, Šumperk, Czech Republic, or from IPK Gatersleben, Germany. *Humulus lupulus* L. DNA was obtained from P. Oriniaková (IPMB, České Budějovice, Czech Republic) and seeds of *Secale cereale* L.

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'Daňkovské' were provided by F. Macháň (ARI, Kroměříž, Czech Republic).

Reagents and DNA manipulations

Enzymes, plasmid vector pBluescript II SK+, *E. coli* strain XL-1 Blue MRF', X-gal, and IPTG were supplied by Stratagene (USA). Hybridization membranes (Hybond-N and N+) were purchased from Amersham (UK). Unless specified otherwise, all DNA and bacteria manipulations were performed according to Ausubel *et al.* (1991). Probes for membrane hybridizations were labelled using AlkPhos Direct labelling and detection system (Amersham) and used according to manufacturer's recommendations. Stringent post-hybridization washes were carried out twice for 10 min at 55 or 60 °C.

Cloning of repetitive genomic DNA

Total genomic DNA was extracted from leaves as described by Dellaporta, Wood and Hicks (1983). Aliquots containing 15 μg of the genomic DNA were digested using different restriction endonucleases and separated on ethidium bromide-stained 2% agarose gels. DNA fragments were isolated from selected bands visible on the gel and cloned into pBluescript II SK+. Colonies obtained after *E. coli* transformation were used for colony hybridization with labelled *V. faba* DNA. Clones showing strong hybridization signals were selected for further analysis.

Dot-blot and Southern hybridizations

For estimation of copy numbers of newly isolated repeats in the field bean genome, serial dilutions of genomic DNA corresponding to $10^2 - 10^5$ copies of haploid *V. faba* genome ($1C = 13{\cdot}5$ pg, 13×10^3 Mbp), were dot-blotted along with $5 \times 10^8 - 2 \times 10^{11}$ copies of respective repetitive sequences. For Southern hybridizations, 3 μg of digested genomic DNA were run on 1% agarose gels and blotted onto the membranes by capillary transfer. Hybridization probes were prepared by labelling cloned DNA fragments amplified from plasmid DNAs using T3 and T7 primers. For generation of probes for *Ty1-copia* and *Bam*HI repeats, PCR was employed with *V. faba* genomic DNA and sequence specific pairs of primers. Fragments of the *Ty1-copia* elements were amplified using primers and reaction conditions described by Flavell *et al.* (1992). The primers for amplification of the 250 bp *Bam*HI repeat (BamHI-F: 5′-CTG GGA GAA TAG CAA GAT G-3′, BamHI-R: 5′-ATC CTT SYT CAG GTC CTT CA-3′) were derived from conserved regions of known sequences (GenBank Acc. Nos. M16847-M16852).

DNA sequencing and sequence analysis

Selected clones were sequenced by the dideoxy chain termination method (Sanger, Nicklen and Coulson, 1977) using the Silver Sequence DNA Sequencing System (Promega, USA). All new sequences were searched for homology to sequences in the GenBank (release 109.0) and EMBL (release 56) databases using BLASTN or BLASTX 2.0.2 (Altschul *et al.*, 1997) or FASTA 3.06 (Pearson and Lipman, 1988). Comparisons between the sequences as well as various sequence analyses (ORF prediction, searches for direct and inverted repeats, secondary structure analysis, homology with tRNA) were performed using PC/Gene 6.60 (IntelliGenetics) and Dot-plot 3.0 (created by Ramin Nakisa, Oxford University) programs.

FISH and combined PRINS-FISH

Cell-cycle synchronization, preparation of squashes from root tips of *V. faba* seedlings, and *in situ* hybridizations were carried out as described previously (Doležel, Číhalíková and Lucretti, 1992; Leitch *et al.*, 1994). Post-hybridization washes included a stringent wash for 10 min in 50% formamide in 2× SSC, followed by a wash in 2× SSC for 10 min, all at 42 °C. Sites of probe hybridization were detected using avidin-Cy3 and the signal was amplified using biotinylated anti-avidin and avidin-Cy3. The chromosomes were counterstained with DAPI (4′,6-diamidino-2-phenylindole) (200 ng ml^{-1} in 4× SSC, 0·2% Tween-20).

In order to determine more precisely the distribution of new sequences on individual chromosome types, a combined PRINS-FISH technique (Kubaláková *et al.*, 1998) was performed with purified chromosome suspensions dried on microscope slides. The PRINS (primed *in situ* DNA labelling; Koch *et al.*, 1989) reaction was done using an oligonucleotide primer specific for the *Fok*I repeat and the reaction mix containing fluorescein-11-dUTP (Macas *et al.*, 1995; Kubaláková, Macas and Doležel, 1997). After completion of the reaction the slides were passed through an ethanol series, hybridized with denatured probe and processed as described above.

The slides were examined with an Olympus BX 60 fluorescence microscope equipped with appropriate filter sets and black and white CCD camera connected to a PC running the ISIS software (Metasystems, USA). The images corresponding to DAPI, FITC (fluorescein isothiocyanate) and Cy3 were superimposed after contrast and background optimization. Signal positions on chromosomes were measured using ISIS software. Six to 28 chromosomes were analysed and mean values were calculated to determine the relative position of each signal.

RESULTS AND DISCUSSION

Field bean genomic DNA was digested with 34 different restriction endonucleases and electrophoresed on 2% agarose gels in order to reveal bands corresponding to DNA fragments containing repetitive sequences. To avoid cloning already known repeats, Southern blots of the digests were prepared and hybridized with labelled probes for *V. faba* rDNA genes (probe Ver17; Yakura and Tanifuji, 1983), the 250 bp subfamily of *Bam*HI repeats (Kato *et al.*, 1985), and part of the *Ty1-copia* element (Pearce *et al.*, 1996). Sixteen bands showing no hybridization with these probes were then isolated from the gel and the DNA was cloned.

TABLE 1. *Characteristics of newly isolated repetitive sequences*

ID	Length (bp)	AT content (%)	Copy number (per 1*C*)	Proportion of *V. faba* genome (%)	Flanking restriction sites	Internal restriction sites (position)	GenBank accession number
TIII15	119	53·8	5×10^5	0·46	*Taq*I	*Fok*I (47)	AJ222866
HII2/7	194	63·8	5×10^4	0·07	*Hinc*II	*Alu*I (66) *Hind*II (61)	AJ222863
RS2/11	203	51·8	7×10^4	0·11	*Rsa*I	*Alu*I (75)	AJ222864
TIV10	233	60·5	10^5	0·18	*Taq*I	*Alu*I (181) *Hind*II (179)	AJ222868
TI33	109	49·5	3×10^5	0·25	*Taq*I	*Alu*I (57)	AJ222865
TIII17	187	58·3	3×10^5	0·43	*Taq*I	*Alu*I (7) *Eco*RI (96) *Hind*III (5)	AJ222867

Resulting colonies were again screened with the probes for known repeats and also with the field bean genomic DNA to confirm their repetitive character. Only these clones that did not hybridize with the known repetitive probes, but that showed strong hybridization signal with the genomic DNA, were selected for detailed analysis. Eleven isolated bands yielded clones which fulfilled these criteria, and a total of 20 clones were fully or partially sequenced.

Computer analysis of data obtained from sequencing revealed that six clones contained novel sequences. They were named according to the enzymes used for the respective restriction digests and identification numbers of bands from which they originated (Table 1). The other fragments were homologous either to some of these clones or to known field bean repetitive sequences which were not used for screening prior to cloning (the *Fok*I repeat, the large spacer of rDNA, or the 1500 or 1750 bp *Bam*HI repeat subfamilies). All new clones represented highly repetitive sequences, having copy numbers from 5×10^4 (HII2/7) to 5×10^5 (TIII15) per 1*C*, which corresponds to 0·07–0·46% of the field bean genome (Table 1). However, none of the new repeats was as abundant as the previous described *Fok*I (Kato *et al*., 1984), *Bam*HI (Kato *et al*., 1985) and *Ty1-copia* (Pearce *et al*., 1996) elements. Since a very thorough screening has been performed, this result suggests that the *V. faba* genome contains only a few families of DNA repeats having individual abundances exceeding 1% of the genome.

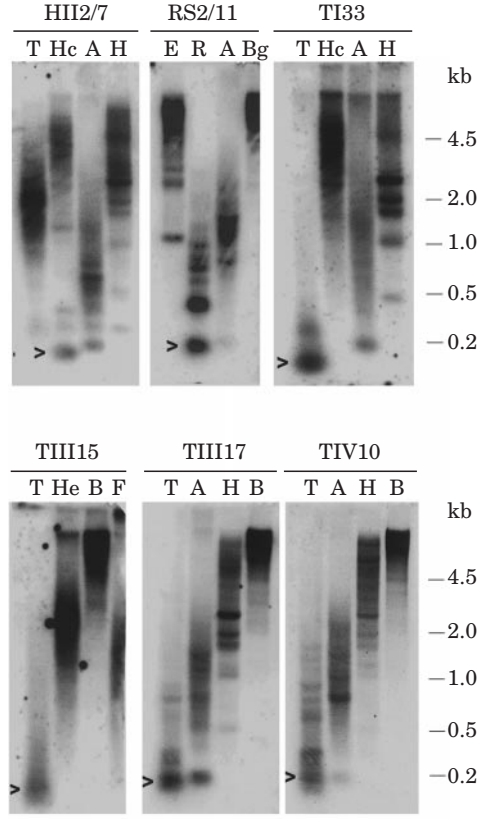


FIG. 1. Genomic organization of new repetitive sequences. Southern blots of field bean genomic DNA digested using various restriction enzymes (H, *Hind*III; Hc, *Hinc*II; T, *Taq*I; A, *Alu*I; R, *Rsa*I; M, *Mbo*I; F, *Fok*I) were probed with the individual repeats. The bands from which the sequences were isolated are marked by arrows. The bands from which full length sequence of TI33 and TIII17 repeats were isolated are marked by asterisks.

Dispersed repeats

Southern hybridizations of blots containing *V. faba* genomic DNA digested with various restriction endonucleases were performed for all six repetitive DNAs (Fig. 1). The hybridization patterns of five of these clones (TIV10, TI33, TIII17, HII2/7, and RS2/11) consisted of strong background smears with only a few prominent bands, suggesting their dispersed organization in the field bean genome. This type of genomic organization was proved by results of fluorescent *in situ* hybridization on mitotic chromosomes showing multiple sites labelled along all six chromosome pairs (Fig. 2A).

There was a notable similarity among the hybridization patterns produced by probing digested genomic DNA with the sequences TI33 and TIII17 (Fig. 1). This finding suggested they represent parts of a larger repetitive element. To isolate full length clone of this repeat, the genomic DNA was digested using *Hind*III and size-fractionated on an agarose gel. DNA fragments of about 600, 1300 and 1800 bp corresponding to the bands showing hybridization

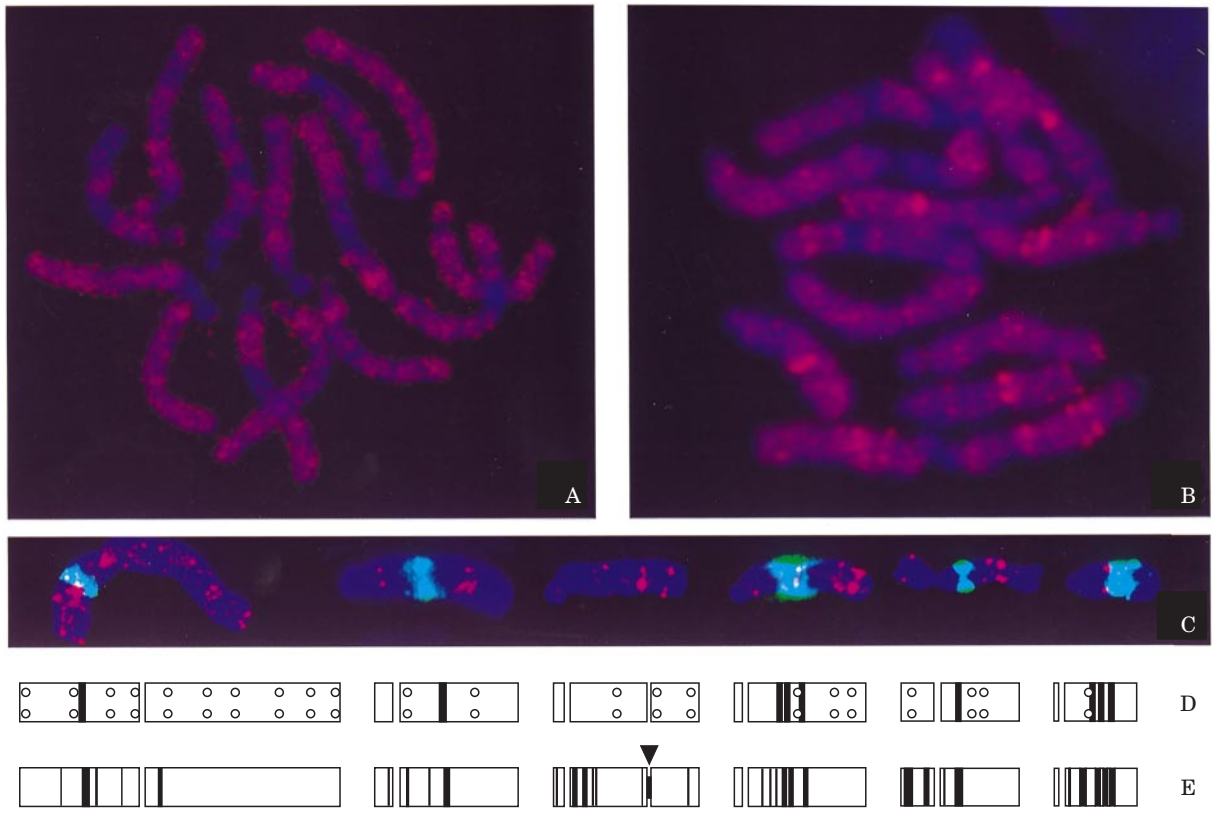


FIG. 2. Localization of repetitive sequences on metaphase chromosomes of *Vicia faba*. *In situ* hybridization using probes for RS2/11 (A) and TIII15 (B) on metaphase plates of field bean with standard (wildtype) karyotype. The probes were labelled by biotin and detected using avidin-Cy3 (red); chromosomes were counterstained with DAPI (blue). C, Simultaneous localization of TIII15 and *Fok*I repeats to chromosomes from the field bean line with reconstructed karyotype ACB. The *Fok*I repeats were labelled by FITC in the PRINS reaction (green) and TIII15 sequences were subsequently visualized by *in situ* hybridization with biotin-labelled probe which was detected using avidin-Cy3 (red). Chromosomes were counterstained with DAPI (blue). D, Schematic representation of the signal positions on chromosomes of the ACB karyotype obtained after simultaneous localization of the TIII15 (circles) and *Fok*I (bands) repeats. E, Scheme of the chromosomes of the ACB karyotype, illustrating Giemsa bands including the NOR (marked by arrowhead), and interstitial bands (according to Döbel, Schubert and Rieger, 1978).

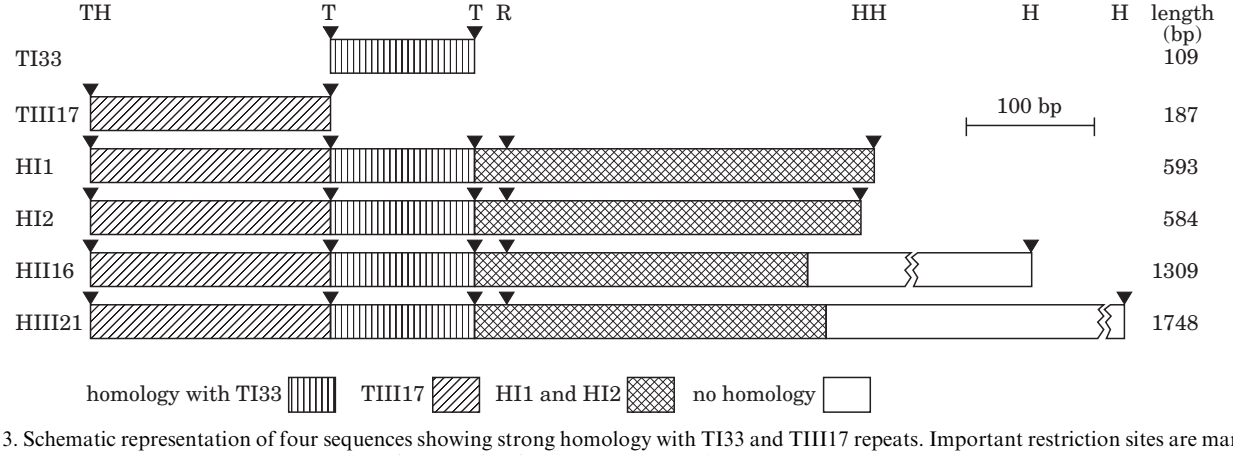


FIG. 3. Schematic representation of four sequences showing strong homology with TI33 and TIII17 repeats. Important restriction sites are marked by arrowheads (T, *Taq*I; H, *Hind*III; R, *Rsa*I).

signals with both sequences were then isolated and cloned. Resulting DNA libraries were successively hybridized with TI33 and TIII17 probes, and the clones showing positive signal with both probes were sequenced. Four clones were analysed in detail and the results are shown in Fig. 3 and Table 2. All of them contained both TI33 and TIII17 sequences arranged in the same orientation, together with an adjacent fragment of about 260 bp. These three regions made up all of the length of the two clones isolated from the 600 bp band (HI1, HI2). They were also conserved in two longer clones HII16 (1309 bp) and HIII21 (1748 bp). Additional sequences present in these clones showed no

TABLE 2. *Clones with homology to both TI33 and TIII17 repeats*

ID	Length (bp)	AT content (%)	Homology with TI33 (%) (position)	Homology with TIII17 (%) (position)	Homology with HI1 (%) (position)	Homology with HI2 (%) (position)	GenBank accession number
HI1	593	58·3	96 (187–286)	91 (1–181)	100 (1–593)	89 (1–538)	AJ011708
HI2	584	59·2	96 (179–287)	93 (1–179)	89 (1–541)	100 (1–584)	AJ011709
HII16	1309	59·7	97 (178–288)	97 (1–178)	86 (1–533)	89 (1–552)	AJ011710
HIII21	1748	65·6	98 (182–288)	95 (1–182)	93 (1–549)	91 (1–562)	AJ011711

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1    TCGAAGTAAC CCTACTATCC AGGTTTCTTC TTACAGCCCT   40

41   ACCATCCAGG TTTCATATCA AGGCAACCCT ACTATCCAGG   80

81   TTCCTTCTCA TAGCCCTACT ATCCAGGCTT TCATCTCGA    119
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Subrepeats :   6   GTAACCCTACTATCCAGG TTTCTTCT   31
              33   AC-G------C------- ----A-A-   58
              63   -C---------------- --C-----   88
              90   A--G--------------C----A---   116
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FIG. 4. Nucleotide sequence of the TIII15 repeat, and comparison of four 26–27 bp long subrepeats within the TIII15. Flanking *Taq*I restriction sites are underlines. The *Taq*I restriction sites between the subrepeats that have possibly been lost by single base deletions are underlined in bold. Dashes denote nucleotide matches.

homology to each other or to other known sequences. Therefore, we presume the fragments of about 600 bp represent the basic elements of this repeat family which may be associated with other genomic sequences.

Dispersed genomic organization is a very common feature of plant repeats (Smyth, 1991; Zhao, Wing and Paterson, 1995; Bennetzen and Kellogg, 1997), especially those which originated from transposable elements (Brandes *et al*., 1997; Galasso *et al*., 1997). However, no homology of the new repeats to known transposons or retrotransposons was found. We also did not find any homology to tRNA genes or an indication of open reading frames or significant secondary structures in these sequences.

Tandem repeat TIII15

Southern hybridization with the clone TIII15 exhibited a 'ladder' hybridization pattern typical for tandem repeats, with the smallest visible band of about 60 bp (Fig. 1). Apart from a weak dispersed signal, FISH with this clone showed preferential hybridization to a few major sites on each chromosome (Fig. 2B). To map these sites more precisely, we simultaneously localized TIII15 with *Fok*I repeats using a combined PRINS-FISH technique. In this procedure, the *Fok*I repeats were first labelled by fluorescein in the PRINS reaction using sequence specific primer, and the chromosomes were then subjected to FISH to visualize the TIII15 sequences. Since the labelling of *Fok*I elements produces characteristic bands at defined positions on field bean chromosomes (Fuchs *et al*., 1994), it allowed us to determine positions of TIII15 signals on individual chromosome pairs. Twenty-two major hybridization sites were reproducibly detected, some of them located near to NOR, telomeric, and centromeric regions (Fig. 2C and D). An interesting finding was the presence of TIII15 signals within the heterochromatic regions containing *Fok*I repeats on chromosomes 1, 4 and 6. However, some signals were also associated with heterochromatic regions lacking *Fok*I sequences as well as with the euchromatin (Fig. 2D and E).

The nucleotide sequence of clone TIII15 contained easily distinguishable internal subrepeats. The sequence organization suggested that TIII15 might originate from four tandemly arranged identical units (Fig. 4). An interesting observation is that the length of the dimer of these units (58 bp) is very close to that of the *Fok*I repeat (59 bp), which also contains two tandem subrepeats. However, the TIII15 and *Fok*I repeats share no sequence homology, suggesting they do not have common origin. Sequence organization of this type is characteristic of 'satellite' repeats (Grellet *et al*., 1986; Ingham *et al*., 1993; Xia,

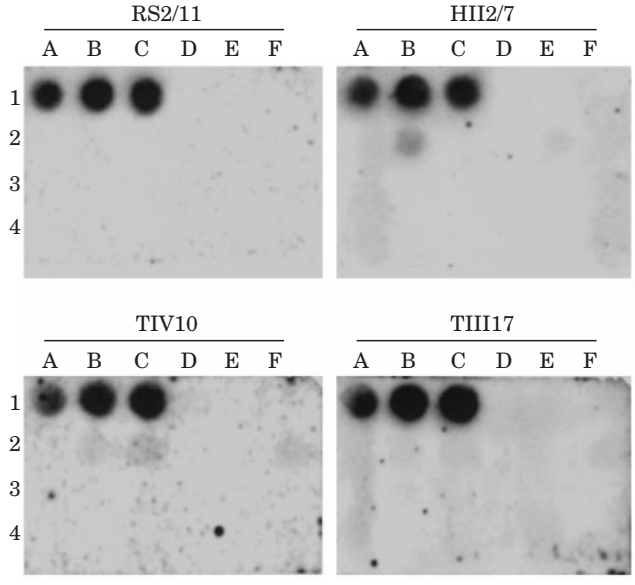


FIG. 5. Determination of the species specificity of the newly isolated repeats. Dot-blots of genomic DNAs from different species were probed using the RS2/11, HII2/7, TIV10 and TIII17 repeats. The sequences TIII15 and TI33 gave the same result as RS2/11 (not shown). The positions of the individual species on the dot-blots were as follows: 1A, *Vicia faba* ssp. *faba* var. *equina* Pers 'Inovec'; 1B, *V. faba* L. ssp. *faba* var *faba*; 1C, *V. faba* ssp. *faba* var. *equina*, Pers.; 1D, *V. melanops* Sibth. & Smith; 1E, *V. narbonensis* L.; 1F, *V. sativa* L.; 2A, *V. lathyroides* L.; 2B, *V. lutea* L.; 2C, *V. michauxii* Sprengel; 2D, *V. panonica* Grantz; 2E, *V. grandiflora* Scop; 2F, *V. peregrina* L.; 3A, *V. hybrida* L.; 3B, *V. villosa* Roth; 3C, *Pisum sativum* L.; 3D, *Phaseolus vulgaris* L.; 3E, *Glycine max* (L.) MERR; 3F, *Lupinus angustifolius* L.; 4A, *Cicer arietinum* L.; 4B, *Vigna unguiculata* (L.) Walp.; 4C, *Vicia sepium* L.; 4D, *Pisum elatius* (M. B.) STEV.; 4E, *Humulus lupulus* L.; 4F, *Secale cereale* L.

Selvaraj and Bertrand, 1993; Galasso *et al*., 1995), which presumably evolved via multiplication of shorter (about 30 bp) sequences (Ingham *et al*., 1993). Typical features of these repeats are their presence exclusively in heterochromatin and their tandem arrangement. Although the tandem arrangement of TIII15 is obvious from Southern blots (Fig. 1) and *in situ* hybridization revealed localization of the repeat in some of the heterochromatic bands and, in some cases, even its simultaneous localization with *Fok*I repeats, it was also found in other chromosomal regions.

Determination of species-specificity

The presence of newly isolated field bean repetitive DNAs in closely related taxa was examined by their hybridization to dot-blots of genomic DNAs from 12 species of the genus *Vicia* and eight other legumes (Fig. 5). The probes hybridized exclusively to *V. faba* DNA (regardless of variety), except for TIV10, TIII17 and HII2/7 which showed weak hybridization to two (HII2/7) or four (TIV10, TIII17) other species. All three sequences hybridized to *V. lutea*; TIV10 and TIII17 hybridized also to *V. michauxii*, *V. melanops* and *V. peregrina*, whereas HII2/7 showed additional hybridization signal with *V. grandiflora* (Fig. 5B–D). According to the taxonomy of the genus *Vicia* (Maxted, 1995), these species belong to sections Hypechusa (*V. melanops* and *V. lutea*), Peregrinae (*V. michauxii* and *V. peregrina*) and Vicia (*V. grandiflora*).

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